



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-G4

System Integration and Architecture

Software Test Plan

Project Title: Pawfolio

Prepared By: Arong, Kylene Jacaban

Project Information

Item	Details
Project Name	Pawfolio
Course	IT342 – System Integration and Architecture
Project Type	Multi-Platform Pet Health Record Management System
Architecture Applied	Vertical Slice Architecture
Prepared By	Kylene Arong
Version	1.0

1. Introduction

1.1 Purpose

This Software Test Plan defines the testing strategy, procedures, test cases, and regression testing activities for the Pawfolio system after applying Vertical Slice Architecture refactoring.

The purpose of this document is to:

- Verify that all implemented system functionalities work correctly after refactoring.
 - Ensure maintainability and scalability of the application.
 - Identify defects, regressions, and broken functionalities.
 - Provide evidence of functional validation through manual and automated testing.
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1.2 Scope

The testing scope covers the following modules:

Web Frontend

- Authentication
- Dashboard
- Breed Explorer
- Health Records

- Notifications
- Profile
- Settings
- Pet Management

Backend

- REST API Endpoints
 - Authentication Services
 - CRUD Operations
 - Database Integration
 - OAuth Login
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1.3 Testing Objectives

The objectives of testing are:

- Validate all functional requirements.
 - Ensure the system behaves correctly after architectural refactoring.
 - Detect regressions introduced during restructuring.
 - Verify API integration and database operations.
 - Confirm frontend and backend compatibility.
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2. Test Strategy

2.1 Testing Types

Testing Type	Description
Functional Testing	Verifies system functions according to requirements
Regression Testing	Ensures old features still work after refactoring
Integration Testing	Verifies frontend-backend communication
UI Testing	Validates interface behavior and responsiveness

API Testing	Tests backend endpoints and CRUD operations
Manual Testing	Performed by interacting directly with the system
Automated Testing	Uses automated scripts and tools where applicable

2.2 Test Environment

Component	Environment
Frontend	React + Vite
Backend	Spring Boot
Database	Supabase PostgreSQL
IDE	VS Code / IntelliJ IDEA
Browser	Google Chrome
Operating System	Windows 11

3. Functional Requirements Coverage

Requirement ID	Functional Requirement	Module
FR-01	User can register an account	Authentication
FR-02	User can log in	Authentication
FR-03	User can log in using Google OAuth	Authentication
FR-04	User can add pets	Pet Management
FR-05	User can edit pet details	Pet Management
FR-06	User can delete pets	Pet Management
FR-07	User can add health records	Health Records

FR-08	User can edit health records	Health Records
FR-09	User can delete health records	Health Records
FR-10	User can upload profile images	Profile
FR-11	User can update profile information	Profile
FR-12	Dashboard displays pet statistics	Dashboard
FR-13	Dashboard displays upcoming vaccines	Dashboard
FR-14	User can search dog breeds	Breed Explorer
FR-15	User can search cat breeds	Breed Explorer
FR-16	User can log out	Settings
FR-17	Notifications are displayed properly	Notifications

4. Test Cases

4.1 Authentication Module

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-AUTH-01	User Registration	Open register page → Enter valid details → Click Register	Account is created successfully	Passed
TC-AUTH-02	User Login	Enter valid credentials → Click Login	User is redirected to dashboard	Passed
TC-AUTH-03	Invalid Login	Enter invalid credentials	Error message is displayed	Passed
TC-AUTH-04	Google OAuth Login	Click Login with Google	User logs in successfully	Passed
TC-AUTH-05	Logout	Click logout button	User session is terminated	Passed

4.2 Pet Management Module

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-PET-01	Add Pet	Fill pet form → Submit	Pet is added successfully	Passed
TC-PET-02	Edit Pet	Modify pet information	Changes are saved	Passed
TC-PET-03	Delete Pet	Click delete button	Pet is removed	Passed
TC-PET-04	View Pet Details	Open pet details page	Correct pet data displayed	Passed

4.3 Health Records Module

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-REC-01	Add Health Record	Enter record details → Save	Record is stored successfully	Passed
TC-REC-02	Edit Health Record	Modify record data	Updated information displayed	Passed
TC-REC-03	Delete Health Record	Remove selected record	Record is deleted	Passed
TC-REC-04	Upcoming Vaccines	Add upcoming vaccine record	Vaccine count updates correctly	Passed

4.4 Profile Module

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-PRO-01	Update Profile Name	Edit profile name → Save	Updated name is displayed	Passed

TC-PRO-02	Upload Profile Image	Upload image file	Profile image updates successfully	Passed
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4.5 Dashboard Module

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-DASH-01	Display Total Pets	Add pets	Dashboard count updates	Passed
TC-DASH-02	Display Records Count	Add records	Dashboard reflects correct count	Passed
TC-DASH-03	Display Vaccine Due Count	Add upcoming vaccine	Dashboard updates correctly	Passed

4.6 Breed Explorer Module

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-BREED-01	Search Dog Breed	Enter dog breed keyword	Matching breeds displayed	Passed
TC-BREED-02	Search Cat Breed	Switch to cats → Search	Matching cat breeds displayed	Passed
TC-BREED-03	Open Breed Details	Click breed card	Breed details modal appears	Passed

5. Test Scripts / Test Procedures

5.1 Login Test Procedure

1. Launch the web application.
2. Navigate to the login page.
3. Enter a valid email and password.

4. Click the Login button.
5. Verify successful redirection to the dashboard.

Expected Result:

User successfully logs into the system.

5.2 Add Pet Procedure

1. Navigate to Dashboard.
2. Click Add Pet.
3. Enter pet information.
4. Submit the form.
5. Verify pet appears in pet list.

Expected Result:

Pet is successfully added and displayed.

5.3 Add Health Record Procedure

1. Open pet details page.
2. Click Add Health Record.
3. Enter vaccine/medical information.
4. Save record.
5. Verify record appears in records list.

Expected Result:

Health record is successfully stored.

6. Automated Test Cases

6.1 Automated Testing Tools

Tool	Purpose
Browser DevTools	Frontend debugging and network testing

6.2 Automated API Tests

Test ID	API Endpoint	Method	Expected Result
API-01	/api/auth/login	POST	Returns successful authentication
API-02	/api/pets	GET	Returns pet list
API-03	/api/pets	POST	Adds new pet
API-04	/api/records	POST	Adds health record
API-05	/api/users/{id}	PUT	Updates user profile

7. Regression Testing Plan

Regression testing was performed after implementing Vertical Slice Architecture refactoring.

The following areas were validated:

- Authentication workflow
- Dashboard functionality
- Pet management
- Health record management
- Breed explorer APIs
- Profile management
- Notification system
- Frontend-backend integration

Results:

All major functional requirements remained operational after refactoring.

8. Risks and Limitations

Risk	Description
API Limitations	Some external APIs do not provide breed images
OAuth Dependencies	Google OAuth requires internet connectivity
Browser Compatibility	Testing primarily performed on Google Chrome

9. Conclusion

The Pawfolio system successfully passed functional and regression testing after applying Vertical Slice Architecture.

The refactoring improved project organization, maintainability, scalability, and modularity while preserving all major system functionalities.

Testing confirmed that frontend and backend modules continue to operate correctly with the updated architecture.